

Deep Reinforcement Learning for Bomberman

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Abstract

Bomberman combines navigation, delayed rewards, and decisions that can place the agent itself in danger. This report discusses the two agents we developed parallelly, both relying on the Double Deep Q-Network (Double DQN) learning algorithm.

1 Theoretical Background and Agent Architecture

1.1 Bomberman as a reinforcement learning problem

At each time step, the agent observes the game, chooses an action, and receives feedback through game events. Its action set is

$$\mathcal{A} = \{\text{UP}, \text{RIGHT}, \text{DOWN}, \text{LEFT}, \text{WAIT}, \text{BOMB}\}.$$

An episode corresponds to a round of play from the agent’s perspective and ends when it dies or the round terminates. The configured board has 17×17 tiles, with a maximum of four agents and 400 steps per round. The aim of the game is to collect coins and eliminate opponents. Coins yield points immediately, whereas placing a bomb can produce a benefit only after its timer expires. The agent must therefore consider future consequences rather than only the next reward.

The usual theoretical model is a Markov decision chain with states, actions, transition probabilities, rewards, and discount factor γ . The state of the game s_t includes the positions of all agents, bomb timers, and revealed coins, as well as any other relevant game information needed to make decisions. To make optimal decisions, the next action should be chosen based on the expected future rewards, considering both immediate and delayed consequences, captured by the action-value function defined as

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[\sum_{k=0}^{T-t-1} \gamma^k r_{t+k+1} \mid s_t = s, a_t = a \right]. \quad (1)$$

Here, r_{t+1} is the learning reward, T is the episode’s terminal time, and π is the policy. A larger discount factor gives delayed outcomes greater weight. For example, a useful bomb placement may deserve a high value even if the agent must first move away and wait for the explosion.

1.2 From Q-learning to deep Q-learning

A deep Q-network (DQN) uses a neural network to approximate the action-value function, allowing it to generalize across similar states and handle large state spaces. For a given input state s the network outputs a vector of action values $Q(s, a, \theta)$ for all possible actions $a \in \mathcal{A}$.

The learning process updates the network parameters θ using stochastic gradient descent

$$\theta_{t+1} = \theta_t + \alpha \left(Y_t^Q - Q(S_t, A_t, \theta_t) \right) \nabla_{\theta} Q(S_t, A_t, \theta_t), \quad (2)$$

where α is the learning rate and Y_t^Q is the target value for the Q-network at time step t defined as

$$Y_t^Q = r_{t+1} + \gamma; \max_a Q(S_{t+1}, a, \theta_t). \quad (3)$$

Previous steps get stored in the *experience replay* buffer, which allows the network to sample minibatches of past transitions for training, reducing the correlation between consecutive updates. Furthermore the target network with parameters θ^- is used to compute the target values, and it is updated every τ steps from the online network’s parameters such that $\theta^- = \theta_t$. The target used in the update then becomes

$$Y_t^Q = r_{t+1} + \gamma \max_a Q(S_{t+1}, a, \theta^-), \quad (4)$$

where θ^- are the parameters of the target network.

1.3 Double DQN: separating selection and evaluation

The maximum in Equation (4) selects and evaluates an action using the same estimates. The issue with this method is that is prone to overestimation of action values, leading to bias in the learned Q-values. Van Hasselt et al. show that this overestimation also occurs with deep networks and propose Double DQN to reduce it [2].

They therefore propose the Double DQN approach, which uses the online network with parameters θ to select the next action and the target network with parameters θ' to evaluate that action. These two networks are then updated independently, by assigning the change to either of the networks randomly during training. Important is that the changing the roles of the online and target networks is done in a symmetric manner. The double DQN target is defined as

$$Y_t^{\text{DoubleDQN}} = r_{t+1} + \gamma Q\left(s_{t+1}, \arg \max_a Q(s_{t+1}, a, \theta_t); \theta'\right), \quad (5)$$

and note that action selection in the $\arg \max$ still uses the online parameter θ_t . The best action of the greedy policy is still estimated using the current values, but the policy networks θ' is used to provide an unbiased evaluation of that action’s value.

In Bomberman, a relevant motivation is to avoid overvaluing risky actions whose apparent payoff is based on inaccurate predictions. The double DQN approach is beneficial in this context, as it helps the agent get more accurate value estimates for risky actions, leading to safer and more effective decision-making.

1.4 Input representation and feature engineering

The function `state_to_features` converts the observation into seven spatial maps and ten scalar values. The board input has shape $7 \times 17 \times 17$ and uses game information directly rather than rendered screen pixels. Table 1 summarizes its channels.

From these standard features, one can construct further informative representations, such as combining bomb locations with the explosion map to a combined **danger map**.

The scalar input consists of the number of rounds, and steps played so far in the game. Additionally it is useful for training to provide information about the agent’s bomb availability, the number of

Channel	Encoding
Walls	Binary map of indestructible walls.
Crates	Binary map of destructible crates.
Coins	Binary map of currently visible coins.
Bombs	Binary map of placed bomb locations.
Explosion map	Integer map of active explosions; the value represents the remaining duration.
Own position	Onehot encoding of the agent’s tile
Opponents	Onehot encoding of each opponent’s tile.

Table 1: Spatial features implemented in `callbacks.py`.

remaining coins, the number of remaining opponents, the current step, and the previous action. Therefore we defined

Channel	Encoding
Bomb availability	Binary value indicating if the agent can place a bomb.
Number of coins normalised	Float32 value representing the number of all collectable coins divided by the total number of coins.
Number of opponents	Integer value representing the number of opponents.
Current step	Integer value representing the current step in the game.
Previous action	One-hot encoding of the previous action taken by the agent.
Own position	Onehot encoding of the agent’s tile
Opponents	Onehot encoding of each opponent’s tile.

Table 2: Scalar features implemented in `callbacks.py`.

By providing the agent with absolute information about the game state, such as the number of remaining coins, opponents, and the current step, the network can make more informed decisions.

1.5 Exploration with an epsilon-greedy policy

The agent implements an ϵ -greedy policy. Let $\mathcal{A}_m(s)$ be the actions retained by its mask. During training, the action is sampled as

$$a_t = \begin{cases} \text{a uniform random action from } \mathcal{A}_m(s_t), & \text{with probability } \epsilon_t, \\ \arg \max_{a \in \mathcal{A}_m(s_t)} Q_\theta(s_t, a), & \text{with probability } 1 - \epsilon_t. \end{cases} \quad (6)$$

Exploration supplies experience with actions that the current network may undervalue, whereas exploitation uses what has already been learned. Both branches respect the mask. The random branch can also select the greedy action, so ϵ_t is the probability of random selection, not necessarily the probability of taking a non-greedy action.

In order to provide sufficient exploration early in training, but gradually reduce it as the agent learns, we use an exponentially decaying epsilon

$$\epsilon_t = \epsilon_{\min} + (\epsilon_{\text{start}} - \epsilon_{\min}) \cdot \exp\left(-\frac{t}{T_{\text{decay}}}\right). \quad (7)$$

It is beneficial to choose $T_{\text{decay}} = \frac{T_{\text{Half-life}}}{\ln 2}$.

During the whole training process, the agent is faced with different training epochs. This introduces varying scenarios, obstacles, and opponents. To provide adequate exploration, it is a good trick to reset epsilon to a middle value at the beginning of each new training epoch, as discussed in 1.6.2.

1.6 Ruehl based agent

1.6.1 Neural network architecture

Given the combination of spatial and scalar features, it is reasonable to use a network architecture that processes these two types of inputs separately before combining them for decision making. The spatial branch contains four convolutional layers with ReLU activations. Dilated convolutions expand the area visible to a filter without increasing its number of weights, while the third convolution reduces the spatial resolution. Table 3 gives the dimensions derived from `Networks.py` with the batch dimension is omitted.

Component	Operation	Output shape
Board input	Seven spatial channels	$7 \times 17 \times 17$
Convolution 1	3×3 , 32 filters, stride 1, padding 1	$32 \times 17 \times 17$
Convolution 2	3×3 , 32 filters, dilation 2, padding 2	$32 \times 17 \times 17$
Convolution 3	3×3 , 32 filters, stride 2, padding 1	$32 \times 9 \times 9$
Convolution 4	3×3 , 32 filters, dilation 2, padding 2	$32 \times 9 \times 9$
Flatten	Flatten the board features	2592
Scalar branch	Linear $10 \rightarrow 32$, ReLU	32
Concatenation	Join spatial and scalar features	2624
Shared layers	Linear $2624 \rightarrow 128 \rightarrow 128$, ReLU after each	128
Value head	Linear $128 \rightarrow 1$	1
Advantage head	Linear $128 \rightarrow 6$	6

Table 3: Architecture of the RUEHL network for the configured board size.

Each output of the last convolution has a theoretical receptive field of 17×17 input tiles, including padding near the boundary. The flattened features subsequently connect to shared fully connected layers.

Complementing the double DQN strategy, one can employ a dueling network architecture to separately estimate state values and action advantages, which can lead to more stable learning and improved policy evaluation. In the dueling DQN architecture, the action-value function is not represented directly by a single network output. Instead, the network decomposes the estimate of $Q(s, a)$ into two components: a state-value function $V_\theta(s)$ and an action-dependent advantage function $A_\theta(s, a)$ [3]. The state-value function represents the overall quality of being in state s , independently of which action is selected, whereas the advantage function describes the relative benefit of choosing a particular action a in that state.

A naive decomposition of the form

$$Q_\theta(s, a) = V_\theta(s) + A_\theta(s, a)$$

is not identifiable. In particular, an arbitrary constant c can be added to the value function and subtracted from all advantage values without changing Q_θ :

$$V_\theta(s) + c + (A_\theta(s, a) - c) = V_\theta(s) + A_\theta(s, a).$$

Consequently, the network would have no unique way of determining which part of the predicted action value should be represented by V_θ and which part should be represented by A_θ .

To remove this ambiguity, the dueling architecture combines the two streams by subtracting the

mean advantage over all available actions,

$$Q_\theta(s, a) = V_\theta(s) + A_\theta(s, a) - \frac{1}{|\mathcal{A}|} \sum_{b \in \mathcal{A}} A_\theta(s, b). \quad (8)$$

This centers the effective advantage values around zero. An important consequence of this parameterization is obtained by averaging $Q_\theta(s, a)$ over all actions:

$$\frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} Q_\theta(s, a) = V_\theta(s).$$

Thus, under this aggregation rule, the value stream represents the mean predicted action value in a state, while the centered advantage stream determines how the individual action values differ from this common baseline. In particular, because both $V_\theta(s)$ and $\bar{A}_\theta(s)$ are independent of the selected action,

$$\arg \max_a Q_\theta(s, a) = \arg \max_a A_\theta(s, a).$$

Hence, the advantage stream determines the relative ranking of actions, whereas the value stream controls the overall level of their Q-values.

The motivation for this decomposition is that in many states the precise choice of action has only a small effect on the expected return. A conventional DQN must nevertheless estimate a separate $Q(s, a)$ for every action. The dueling architecture allows the network to learn features that characterize whether a state is generally favorable or unfavorable through $V_\theta(s)$, while using the advantage stream only to represent differences between actions. This can make value estimation more efficient in states where several actions have similar consequences [3].

The two quantities are not trained using separate value and advantage targets. Instead, they are combined according to Eq. (8) to obtain $Q_\theta(s, a)$, and the usual temporal-difference loss is applied to this resulting Q-value. Gradients then propagate through the aggregation operation into both the value and advantage streams. The decomposition is therefore an architectural parameterization of the Q-function rather than the introduction of two independently supervised learning objectives.

1.6.2 Training curriculum

The training progression considered in this project starts with navigation, then introduces bomb use, and finally adds opponents. This curriculum is motivated by the difficulty of discovering all three behaviors at once: an inexperienced agent can otherwise die before collecting useful examples of delayed rewards. Table 4 links the progression to the available scenarios.

Stage 1: coin collection. In the `coin-heaven` scenario, visible coins give a relatively direct objective: move through the maze and collect them efficiently. The agent can learn spatial relationships between its position, walls, and rewards. For this initial stage of training, we started the epsilon greedy strategy with $\epsilon = 1$, ensuring maximum exploration. Via reward shaping, the agent receives immediate feedback for collecting coins, and negative feedback for undesirable actions, guiding it towards efficient coin collection.

Scenario	Crate density	Coins	Role in the curriculum
<code>coin-heaven</code>	0	50	Learn movement and collection; the initial stage is intended to run without opponents.
<code>loot-crate</code>	0.75	50	Learn to open routes, reveal coins, and escape after placing bombs.
<code>classic</code>	0.75	9	Combine collection and bomb use with competition against opponents.

Table 4: Configured scenarios and their intended training roles. Crate density is the placement probability before walls and starting areas are cleared. Opponents are configured separately from the scenario.

Stage 2: crate destruction. The `loot-crate` scenario adds destructible obstacles and places coins preferentially underneath them. Successful collection can now require a sequence of actions: approach a crate, place a bomb, reach safety, and return to collect a revealed coin. This stage introduces delayed credit assignment and the conflict between making progress and avoiding self-destruction. Initializing from the pretrained Network on `coin-heaven`, we continued training with substantial epsilon reheating $\epsilon = 0.85$ to encourage exploration in the presence of destructible crates. For this stage, a different reward profile is used, emphasizing successful crate destruction and safe bomb placement, while still rewarding coin collection and very importantly, penalises suicidal actions.

Stage 3: opponents and competitive play. The main goal for this stage is to adapt to less placed coins, while dealing with dynamic opponents. A progression from simpler opponents to stronger ones would let the agent adapt these skills gradually. For this purpose, we iteratively improved the model, by transferring the pretrained network from `loot-crate` to `classic` with slight epsilon reheating $\epsilon = 0.20$, then gradually increased the difficulty of opponents.

1.6.3 Action masking and hand-designed guidance

The mask is an important part of an effective policy. It first rejects movement into walls, crates, bombs, or opponents. It then filters movement and waiting on active explosions or blast tiles of bombs whose timers are at most one. Bomb placement is allowed only when a bomb is available, its blast could hit a crate or opponent, and a static breadth-first search finds a tile outside its blast within the configured bomb timer.

If no action survives the danger filter, the code falls back to the earlier legal-action mask. If that is also empty, the model chooses `WAIT`. This is therefore a heuristic filter, not a guarantee of survival. Its static escape search does not predict moving opponents or jointly simulate other bombs over time.

This guidance reduces the exploration burden, but it also contributes to the observed behavior independently of learned Q-values. Results should therefore describe the complete network-plus-mask policy. The Double DQN update in `train.py` also applies the next-state action mask before the online network selects its maximizing action. The target network then evaluates that selected action.

1.6.4 Reward shaping

The main goal of the agent is to maximize game score at the end of the game, by collecting coins, eliminating opponents, given by

$$S = N_{\text{coins}} + 5N_{\text{opponents killed}}. \quad (9)$$

However, starting from these sparse and delayed objectives alone makes it difficult for the agent to learn effective strategies. Reward shaping addresses this problem, by giving intermediate rewards and punishments that guide the agent’s behavior before the final game outcome is known. For the `ruehl_based_agent`, we rejected anisotropic rewards, meaning rewards that vary depending on the direction or context of the action. Combined with the CNN architecture, this design choice ensures that the agent exploits the symmetry of the game environment. Furthermore, we only introduce additional shaping rewards that are deductible from the input game state, e.g. rewarding the survival of own placed bombs or penalizing the agent for being in the blast radius of an active explosion. Furthermore, one can differentiate between main rewards, connecting to significant game events, and shaping rewards, which serve as fine tuning of the agent’s behavior. The shaping rewards are an order of magnitude smaller than the main rewards.

Main rewards The main contribution, stored under `major`, assigns values to selected game events. Collecting a coin provides a baseline reward, typically 1 point, in accordance with the actual game score. Other rewards will be scaled relative to this baseline. A separate Self-destruction penalty is useful in competitive play. This expresses the trade-off between gaining points and retaining the opportunity to act. Penalizing only death would encourage excessive caution, meanwhile rewarding only attacks would provide insufficient feedback about their danger. Their combination is intended to encourage productive risks with an escape plan.

Shaping penalties The separate shaping terms discourage waiting, entering an active explosion, and, in the competitive profile, immediately reversing movement direction. Their purpose is to discourage inactivity and repeated back-and-forth motion without making these actions impossible. Waiting for an explosion to clear or reversing direction to escape can be necessary, so these penalties should remain small relative to important outcomes.

1.6.5 Task-specific reward profiles

We identified three substantially different challenges, for which we designed corresponding reward profiles: `coin`, `loot_exploration`, and `killer`. Determining the appropriate distribution of rewards was a challenging task that required long trial and error processes. The current values are summarized in Table 5.

Coin collection. The `coin` profile rewards collection and discourages waiting through both a shaping penalty and a small step cost. It gives no direct incentive for crate destruction or bomb placement, focusing the initial task on navigation. Opponent-kill rewards are inherited from the shared reward table but have no effect in a scenario without opponents.

Crate exploration. The `loot_exploration` profile rewards destroying crates, revealing coins, and placing bombs whose blast reaches a crate. These intermediate rewards provide feedback before the agent eventually collects a coin. Removing the step cost and reducing the waiting penalty allows more time for bomb placement and escape sequences, while the self-destruction penalty retains an incentive to survive.

Event or parameter	coin	loot_exploration	killer
Coin collected	1.0	1.0	1.0
Coin revealed	0.1	0.5	0.5
Crate destroyed	0	0.35	0.15
Bomb placed to hit a crate	0	0.05	0.02
Bomb placed to threaten an opponent	0	0	0.75
Opponent killed	6.5	6.5	6.0
Killed by another agent	-1.0	-1.0	-2.0
Self-destruction	-2.0	-2.0	-6.0
Bomb exploded	0	0	0.15
Round survived	0	0	0.75
Waiting (shaping)	-0.08	-0.01	-0.05
Danger (shaping)	-0.02	-0.04	-0.10
Reversing direction (shaping)	0	0	-0.03
Shaping limit c	0.05	0.05	0.15
Step cost λ	0.008	0	0.002

Table 5: Reward profiles in `train.py`. Event values are summed per transition; only the combined shaping contribution is clipped. A zero denotes an event with no assigned reward in that profile.

Competitive play. The `killer` profile emphasizes successful attacks and survival. A bomb that currently threatens an opponent earns an immediate reward of 0.75, while a confirmed kill earns 6.0. Self-destruction costs 6.0, substantially more than death caused by another agent. Smaller crate rewards preserve an incentive to open routes, and surviving the round earns 0.75. The bomb-placement event checks the current blast geometry, not whether the opponent will actually be hit. If both opponents and crates are in range, only the opponent-targeting placement event is added.

1.6.6 Optimization and training state

The training loop stores transitions in a replay buffer of fixed size, and samples minibatches uniformly without replacement, starting after the warm-up period. It implements the Double DQN target from Equation (5): the masked online network selects the next action, and the target network evaluates it. For optimization, the `ruehl_based_agent` uses AdamW with a smooth L1 loss, and clips the gradient norm to 5.0. The target network receives a hard copy of the online weights every `TARGET_UPDATE` gradient updates.

The replay capacity, batch size, discount factor, learning rate, warm-up, update intervals, and evaluation budget are read from `Hyperparams.prm`.

1.6.7 Evaluation and model selection

After `EVAL EVERY TRAINING ROUNDS` training rounds, the agent plays `EVAL ROUNDS` greedy evaluation rounds with $\epsilon = 0$. During these rounds, no transitions are added to the replay buffer and no optimization or exploration decay steps occur. Periodic evaluation ensures that model selection is based on up-to-date performance rather than outdated checkpoints, especially for DQN agents [1]. For coin collection and crate exploration, the selection is based on the actual game score rather than accumulated shaping rewards. In competitive play, the selection score is extended by penalising self-destruction and favoring eliminating opponents.

1.7 The Final Agent

It should be noted that the final agent is only final by name and was not the submitted agent for the competition.

The final agent is based on the same DQN architecture and training curriculum as the Ruehl based agent, but with some modifications to the input representation and reward shaping. The intention behind the final agent was to explore the impact of a bigger Network architecture, and to investigate the effect of a different reward shaping strategy and the use of a more guided action mask on the agent’s performance.

Compared to the Ruehl based agent, the final agent uses a larger network with more convolutional layers and filters, allowing it to capture more complex spatial relationships in the game environment. Additionally, the final agent incorporates a more sophisticated reward shaping strategy that takes into account not only the immediate rewards but also the long-term consequences of actions, encouraging the agent to develop more strategic behaviors. It also incorporates a more guided action mask that restricts the agent’s actions based on the current game state, helping it to avoid dangerous situations and make more informed decisions.

1.7.1 The Features

The features used for this Network are one of the main differences between the Ruehl based agent and the final agent.

The spatial input consists of fourteen 17×17 feature maps containing information such as walls, crates, coins, agent positions, and bomb-related danger. Each one of these maps is designed to capture only one spatial aspect of the game state, which resulted in 14 different channels of information which are summarized in Table 6. The scalar input consists of twenty-two features,

Channel	Feature	Encoding
0	Walls	Binary map of indestructible walls.
1	Crates	Binary map of destructible crates.
2	Coins	Binary map of visible coins.
3	Own position	One-hot encoding of the agent’s current tile.
4	Useful bomb positions	Binary map of free tiles where placing a bomb could hit a crate.
5	Bomb timers	Remaining timer divided by 4 at each bomb’s position; zero elsewhere.
6	Bomb danger	Binary map of tiles within the blast area of any existing bomb, regardless of its timer.
7	Active explosions	Binary map indicating where <code>explosion_map</code> > 0.
8	Free terrain	Binary map indicating where <code>field</code> = 0; tiles may still contain bombs or agents.
9	Opponent positions	Binary map of opponents’ current tiles.
10	Explosion urgency	$1 - t_{\min}/4$ at threatened tiles, where $t_{\min}$ is the smallest timer among bombs whose blasts cover that tile; zero elsewhere.
11	Safe neighboring tiles	Binary map of adjacent tiles for which the survival search finds a surviving continuation after moving there.
12	Own hypothetical blast	Binary map of tiles affected by a hypothetical bomb at the agent’s current position.
13	Opponents’ hypothetical blasts	Binary map of tiles affected by hypothetical bombs at opponents’ current positions.

Table 6: The final agent’s fourteen spatial input channels. Each channel has dimensions 17×17 . Channel indices follow the implementation in `callbacks.py`.

which are summarized in Table 7. They give the network scalar information that break isotropy

which is one of the biggest differences between the final agent and the Ruehl based agent. This was done to guide the agent more during the training process and to achieve a faster convergence of the training. This was done despite the risk of overfitting and will be later discussed in the conclusions section.

Index	Feature	Encoding
0	Coin direction, x	Suggested next step's x -offset toward a coin, divided by 17.
1	Coin direction, y	Suggested next step's y -offset toward a coin, divided by 17.
2	Crate-target direction, x	Suggested next step's x -offset toward a useful bomb position, divided by 17.
3	Crate-target direction, y	Suggested next step's y -offset toward a useful bomb position, divided by 17.
4	Crate within bomb range	Binary indicator that a bomb at the current position could hit a crate.
5	Bomb availability	Integer value of <code>bombs_left</code> .
6	Current danger	Binary indicator that the current tile lies in an existing bomb's blast area or an active explosion.
7	Safe movement available	Binary indicator that at least one movement action has a surviving continuation.
8	Previous position, x	$(x_{\text{previous}} - x)/17$.
9	Previous position, y	$(y_{\text{previous}} - y)/17$.
10	Two positions back, x	$(x_{\text{two back}} - x)/17$.
11	Two positions back, y	$(y_{\text{two back}} - y)/17$.
12	Round progress	Current step divided by 400.
13	Closest opponent distance	Minimum Manhattan distance to an opponent, divided by 17; set to 2 if no opponents remain.
14	Opponent within bomb range	Binary indicator that a bomb at the current position could hit an opponent.
15	Survival after bomb placement	Binary indicator that the search finds a surviving path after placing a bomb and remaining on the current tile for that action.
16	Potential attacking opportunity	Binary conjunction of opponent within bomb range and survival after bomb placement.
17	Predicted trapped opponent	Binary indicator that an opponent lies in the proposed bomb's blast area and the search finds no escape for that opponent.
18	Opponent direction, x	Suggested next step's x -offset toward an opponent, divided by 17.
19	Opponent direction, y	Suggested next step's y -offset toward an opponent, divided by 17.
20	Remaining opponents	Number of opponents divided by 3.
21	Current score	Raw game score without normalization.

Table 7: The final agent's twenty-two scalar input features. Here, (x, y) is the agent's current position. Direction features describe the next step suggested by the path search. Historical positions refer to recorded position changes, with consecutive repetitions omitted; missing history entries are encoded as zero. Indices follow the implementation in `callbacks.py`.

1.7.2 Neural Network Architecture

The final agent's network architecture is similar to the Ruehl based agent, but with some modifications to the convolutional layers and the fully connected layers.

It is based on the same two headed logic, where a CNN evaluates the spatial features and a fully connected network evaluates the scalar features. The outputs of both networks are then concatenated and passed through a series of fully connected layers to produce the final Q-values for each action. In theory a simple CNN should be sufficient for this task, but we decided to add a fully connected network to potentially speed up the training process.

Simply increasing the number of layers in the fully connected network and the CNN lead to a significant increase in the number of parameters, which slowed down the training process and skewed the ratio of spatial to scalar features which resulted in a worse performance of the final agent. Another problem with that simple approach was the increase in dead neurons in the fully connected layers and the CNN had the problem that after compressing the resolution in each layer, the network became worse in seeing objects at a distance.

After testing this simple approach, we decided to further increase the complexity of the so called **HybridDQN**. The final agent’s HybridDQN combines a convolutional neural network with a

Component	Operation	Output shape
Board input	Fourteen spatial channels	$14 \times 17 \times 17$
Convolution 1	3×3 , 32 filters, stride 1; GN, SiLU	$32 \times 17 \times 17$
Residual block 1	Two 3×3 convolutions, 32 filters; skip connection	$32 \times 17 \times 17$
Convolution 2	3×3 , 64 filters, stride 2; GN, SiLU	$64 \times 9 \times 9$
Residual block 2	Two 3×3 convolutions, 64 filters; skip connection	$64 \times 9 \times 9$
Convolution 3	3×3 , 96 filters, stride 2; GN, SiLU	$96 \times 5 \times 5$
Residual block 3	Two 3×3 convolutions, 96 filters; skip connection	$96 \times 5 \times 5$
Flatten	Flatten the spatial features	2400
Scalar input	Twenty-two engineered features	22
Scalar branch	Linear $22 \rightarrow 64 \rightarrow 64$, LN and SiLU after each layer	64
Concatenation	Join spatial and scalar features	2464
Shared layers	Linear $2464 \rightarrow 512 \rightarrow 256$, LN and SiLU after each layer	256
Value head	Linear $256 \rightarrow 128 \rightarrow 1$, SiLU after the hidden layer	1
Advantage head	Linear $256 \rightarrow 128 \rightarrow 6$, SiLU after the hidden layer	6
Dueling output	$Q(s, a) = V(s) + A(s, a) - \frac{1}{6} \sum_{b=1}^6 A(s, b)$	6

Table 8: Architecture of the final agent’s **HybridDQN**. The batch dimension is omitted. GN denotes group normalization with eight groups; LN denotes layer normalization. All convolutions use padding 1. Each residual block uses stride 1 and applies convolution–GN–SiLU–convolution–GN, followed by addition of the block input and a final SiLU activation.

fully connected network to process two complementary representations of the game state. The convolutional branch extracts spatial patterns from the board, while the fully connected branch processes engineered features that describe relevant game situations explicitly. Their outputs are combined and passed to a dueling architecture, which estimates a shared state value and the relative advantages of individual actions.

The first convolution transforms the 14 spatial feature maps into 32 feature maps while preserving their spatial dimensions. Two subsequent convolutional stages increase the number of channels to 64 and 96, respectively. These stages use a stride of two, reducing the spatial resolution first to 9×9 and then to 5×5 . This lets the network combine information across increasingly large regions of the board while limiting the number of features passed to the fully connected layers. Each of these convolutions is followed by group normalization with eight groups and a SiLU activation. The SiLU function was chosen, because with ReLU we previously observed a significant number of dead neurons in the fully connected layers, which we hoped to prevent with the SiLU activation.

Each convolutional stage also contains a residual block. A residual block processes its input through two additional convolutions and then adds the original input to their output. For an input tensor x , the block computes

$$ResBlock(x) = GN_2(Conv_2(SiLU(GN_1(Conv_1(x))))), \quad (10)$$

followed by

$$y = SiLU(x + ResBlock(x)). \quad (11)$$

Both convolutions use 3×3 kernels, stride one, and padding one. They preserve the number of channels and the spatial dimensions, so the input x and the transformed features $F(x)$ can be added element by element. Consequently, the residual blocks preserve the shapes $32 \times 17 \times 17$, $64 \times 9 \times 9$, and $96 \times 5 \times 5$, respectively. The resolution changes occur in the strided convolutions between the blocks.

The addition of x to the output of the residual block allows the convolutional path to learn corrections or additions to an existing representation. For example, a block can refine features describing nearby crates by incorporating surrounding escape routes. During backpropagation, the addition also provides a direct path for gradients alongside the path through the convolutional layers, making it easier to train a deeper spatial encoder. This was implemented to allow the network to learn from a pretrained starting point, and to improve the learning of more complex spatial relationships.

The fully connected branch takes the 22 engineered features as input and processes them through two linear layers with 64 hidden units each. Each layer is followed by layer normalization and a SiLU activation. The output of this branch is a 64-dimensional feature vector that captures relevant scalar information about the game state.

The final agent uses the dueling decomposition described for the Ruehl based agent, combining its value and advantage outputs according to Equation (8).

In the final agent, both heads receive the shared 256-dimensional representation. Each head contains a separate fully connected hidden layer with 128 units and a SiLU activation. A final linear layer produces one value estimate in the value head and six action-specific outputs in the advantage head.

Compared with the direct output layers of the Ruehl network, these additional hidden layers allow each head to apply its own nonlinear transformation to the shared representation before producing its output. The aggregation rule remains unchanged.

This architecture allows the network to be deeper and more complex, compared to the Ruehl based agent, while solving the problems of the simple approach of just increasing the number of layers in the fully connected network and the CNN.

1.7.3 The Action Mask

One of the main differences between the Ruehl based agent and the final agent is the use of a more guided action selection through the action mask. The intent was to reduce the amount of possible scenarios to only the ones that were legal moves and that did not lead to an immediate suicide of the agent. The reasoning behind that restrictive approach, was to reduce the training time.

The final agent uses a hand-designed action mask to restrict action selection according to physical feasibility, predicted survival, and the usefulness of bomb placement. For each state, the mask assigns a Boolean value to each of the six actions.

The survival check performs a breadth-first search, which was implemented from the rule based agent, over positions and future time steps, using a horizon of six steps. The first action is fixed to the candidate being evaluated; subsequent steps may contain any of the four movement actions or waiting. A candidate passes the check if at least one continuation survives the entire search horizon. The search accounts for walls, crates, bomb occupancy, bomb timers, and active explosions. Consequently, it can reject a move that is safe immediately but leads to an unavoidable explosion a few steps later.

For bomb placement, the search inserts a hypothetical bomb at the agent’s current position. The

agent remains on that tile during the placement action, and the search then checks whether a surviving continuation exists. Bomb placement additionally requires an available bomb and a blast area that could reach at least one crate or opponent. Table 9 summarizes the rules.

Action or stage	Rule
Movement	Allow the action if the destination is traversable and the survival search finds a continuation lasting the full six-step horizon.
WAIT	Allow waiting if remaining on the current tile for the first step permits a surviving continuation.
BOMB	Allow placement if a bomb is available, its blast could reach a crate or opponent, and the survival search succeeds with the additional bomb included.
First fallback	If the preceding rules reject every action, allow movement to adjacent tiles inside the board that contain no wall, crate, bomb, or opponent. Explosion danger is disregarded at this stage; waiting and bomb placement remain excluded.
Final fallback	If no movement action passes the first fallback, allow WAIT , even if survival cannot be established.

Table 9: Action-mask rules of the final agent. The fallback stages are applied only when the preceding stage leaves no permitted action.

The mask is applied during both exploration and exploitation. Exploration samples uniformly from the permitted actions. For greedy selection, excluded actions receive a temporary value of $-\infty$:

$$\tilde{Q}(s, a) = \begin{cases} Q(s, a), & \text{if the mask permits } a, \\ -\infty, & \text{otherwise.} \end{cases} \quad (12)$$

The agent then selects the action maximizing $\tilde{Q}(s, a)$. During Double DQN training, the next-state mask also restricts the online network’s action selection when constructing the bootstrap target; the target network evaluates the selected action.

The survival prediction uses a simplified model: terrain remains fixed during the search, opponents’ current positions block only the first step, and future opponent movements and new bomb placements are not predicted. Together with the fallback rules, these assumptions mean that the mask cannot guarantee survival. It provides explicit guidance that contributes to the agent’s behavior alongside the learned Q-values, while restricting exploration to actions admitted by its rules.

1.7.4 The Reward Shaping

One principle that guided the design of the final agent’s reward shaping was the rule, that nothing could be rewarded or punished that was not directly observable for the agent, i.e the information that is being rewarded must be part of the input features of the network.

The final agent combines rewards for game outcomes with smaller rewards for intermediate behavior. These intermediate signals provide feedback before an action sequence produces a measurable outcome. For example, destroying a crate and collecting the coin underneath requires the agent to approach a suitable position, place a bomb, escape its blast, and return after the explosion. Reward shaping provides feedback at several points in this sequence.

The learning reward is constructed by summing the contributions of the events associated with a transition. Multiple contributions can therefore occur simultaneously: collecting a coin can coincide with movement-related rewards and the step penalty. Events without an assigned reward contribute zero. The current reward configuration is summarized in Table 10.

Event or condition	Reward
<i>Game outcomes</i>	
Opponent eliminated	+25.00
Coin collected	+7.00
Round survived	+1.00
Crate destroyed	+0.10
Coin revealed	+0.05
Suicide	-20.00
Got killed by an opponent	-15.00
<i>Bomb placement and escape</i>	
Bomb predicted to trap an opponent	+0.75
Bomb threatening an opponent without a predicted trap	+0.30
Bomb targeting a crate without threatening an opponent	+0.20
Bomb placement failing the preceding checks	-0.50
Leaving the tracked own bomb's blast area	+0.30
Remaining in that blast area without a predicted survival path	-1.00
<i>Movement guidance</i>	
Moving closer to a coin	+0.05
Moving farther from a coin	-0.05
Moving closer to an opponent	+0.05
Moving farther from an opponent	-0.05
Moving closer to a useful crate-bombing position	+0.03
Moving farther from a useful crate-bombing position	-0.03
<i>Behavioral penalties</i>	
Invalid action	-1.00
Waiting	-0.05
Returning to the position recorded two steps earlier	-0.50
Step cost on regular training transitions	-0.01

Table 10: Reward configuration of the final agent. Positive bomb-placement rewards require a predicted survival path. Bomb-placement categories are mutually exclusive, and the self-destruction penalty replaces the ordinary death penalty.

Collecting coins and eliminating opponents receive substantially larger rewards than individual shaping events. This gives completed objectives greater weight than intermediate progress. Crate destruction and coin revelation receive smaller rewards because they create opportunities for later collection. These values define a learning objective with its own relative priorities: an opponent elimination is worth approximately $25/7 \approx 3.57$ collected coins.

Death receives a substantial negative contribution. The implementation distinguishes self-destruction, penalized by -20 , from other deaths, penalized by -15 . Since self-destruction also generates the general death event, the reward function explicitly applies only the self-destruction penalty in this case. This prevents an unintended combined penalty of -35 . Other rewards generated during the same transition still contribute to the total.

Bomb-placement shaping distinguishes between predicted usefulness and confirmed success. A placement receives $+0.75$ if the agent has a predicted escape route and an opponent within the blast area is predicted to have no escape. If an opponent is threatened but no trap is predicted, the reward is $+0.30$. A placement targeting a crate receives $+0.20$ when an escape is predicted and no opponent is threatened. These conditions are evaluated in priority order, so a single placement receives only one placement classification. Placements failing these checks receive -0.50 . The larger reward for an actual elimination is awarded separately when the outcome occurs.

The agent also tracks its most recent bomb placement. Leaving that bomb’s blast area produces a reward of $+0.30$. Remaining inside the area produces a penalty of -1.00 only when the survival search finds no surviving continuation. This condition allows temporary movement within the blast area while an escape remains possible.

Movement shaping uses changes in Euclidean distance to coins, opponents, and positions from which a bomb could destroy a crate. Decreasing the distance produces a small positive reward, while increasing it produces a negative contribution. The crate-related target is therefore a useful placement position rather than the crate’s occupied tile. These signals provide frequent directional feedback, although Euclidean distance does not account for obstacles or the length of a traversable route. When constructing therewards for the movements it is necessary to make sure that they aren’t asymmetrical. If for example moving closer to a coin would be rewarded with $+0.05$, but moving away from a coin would be punished with -0.04 , the agent could simply move back and forth to collect a reward of $+0.01$, which would teach the agent oscillatory behaviour.

Waiting and invalid actions receive penalties, and regular training transitions include a small step cost. The position-history check also penalizes a return to the position recorded two steps earlier. This discourages back-and-forth movement, but the same condition can also be satisfied by repeated waiting.

As an illustrative example, a sequence involving a predicted-safe crate-targeting placement, leaving the blast area, destroying one crate, revealing one coin, and collecting that coin contributes

$$0.20 + 0.30 + 0.10 + 0.05 + 7.00 = 7.65.$$

This is the undiscounted sum of those event contributions; movement rewards, step costs, and other events would modify the sequence’s total reward.

The shaping terms also introduce trade-offs. A necessary detour may increase the distance to a target, and retreating from an opponent may be essential for survival. Waiting can likewise be appropriate while an explosion clears. Furthermore, repeated shaping rewards can accumulate because their combined contribution is not clipped. Evaluation should therefore examine game score, coin collection, eliminations, and survival alongside learning reward. Controlled comparisons with individual shaping terms removed would be needed to establish which components improve performance.

2 Results and Discussion

2.1 Ruehl based agent

Table 11 summarizes the best greedy evaluation observed during each curriculum stage. Each reported score is the mean over the 20 evaluation games used for checkpoint selection, whereas the first row gives the number of training games assigned to the stage.

Metric	Coin collection	Loot crate	Peaceful killer	Classic killer	Classic self-play	Classic hard opponents
Training games	30,000	60,000	120,000	30,000	30,000	18,500
Mean game score	50.00	44.90	6.45	7.95	7.10	3.95
Mean coins collected	50.00	44.90	3.20	4.20	2.60	1.95
Mean opponents killed	0.00	0.00	0.65	0.75	0.90	0.40
Completion rate (%)	100	20	0	0	0	0
Suicide rate (%)	0	5	20	5	10	10
Mean crates destroyed	0.00	43.40	44.05	44.60	36.35	29.10

Table 11: Best-model evaluation metrics across the six training stages. Values are taken from the corresponding `Hyperparams.prm` snapshots.

Figure 1 depicts how the selected best-model score evolves with training. The evaluations take place every 200 games, and the current best model is saved if its score improves. The score converges fairly quickly in each stage, approximately within the first 5,000 games, and then plateaus. In the `classic killer` stage, the score has jumped at 28,000 games, suggesting that improvement continues beyond the initial plateau. This slow progression is characteristic of DQN training: periodic target-network synchronization changes the temporal-difference targets and therefore the loss landscape, while evaluation only measures the resulting policy. The results after a long enough training period are summarized in Table 11.

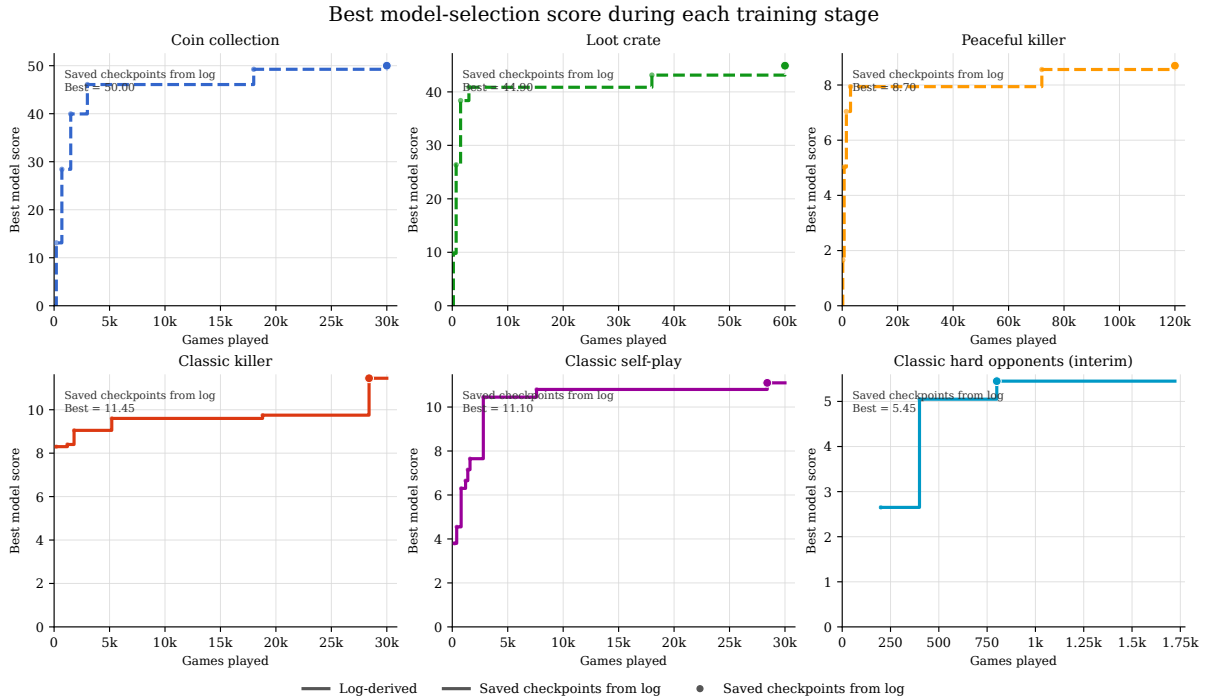


Figure 1: Best model-selection score as a function of games played in each training stage. The evaluations take place every 200 games, and the current best model is saved if its score improves. The scores of the best models at each checkpoint are reported in Table 11.

2.2 Final agent

The final agent was not the submitted agent for the competition, but was trained in parallel to the Ruehl based agent. Despite that it did not achieve the same level of performance as the Ruehl based agent. This will be demonstrated without showing the results of several thousand games, but rather a short version of it.

The following two plots show the performance of the final agent as it was trained on 300 games of loot crate with 50 coins without an opponent, followed by around 600 games of classic with nine coins and with three rule based agents as opponents. In order to determine whether or not a completely new network would be able to improve, the rolling average of the coins collected, the opponents killed, the crates destroyed, the score, the suicide ratio and the steps until the game ends were calculated over the last 10 games. Figure 2 shows these plots.

As can be seen, during the training in loot crate, the network made constant improvements over the duration of the 300 rounds. The plots showing the classic games, however show that the network had significant trouble adapting to a much more complex training environment. Despite double the training and the previous knowledge, from the loot crate training, the score only shows a very slight upwards trend and no improvement in the kill-ability.

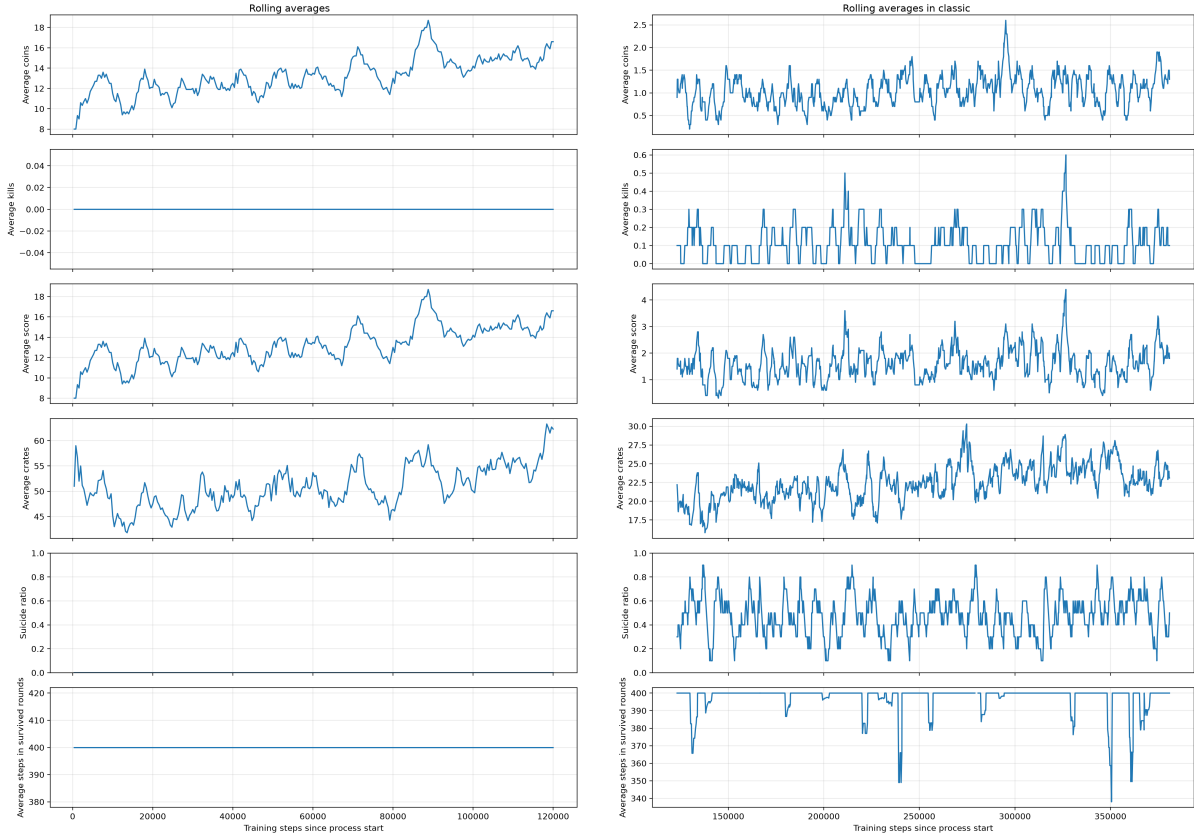


Figure 2: Learning curves of the final agent: loot-crate without opponents (left) and classic with three rule-based opponents (right).

Comparing 2 to 1 a very clear difference is the amount of training rounds needed to see improvements in the score. While the Ruehl based agent needed several thousand games to properly train, the final agent showed significant improvements in the score after only 300 rounds. This supports the idea behind the approach of the final agent, where a more guided approach to the action selection and a bigger neural network should result in a faster convergence of the network.

However, it should be noted that the final agents training time per game was significantly longer than the Ruehl based agent where it crossed the 60 second mark in several training rounds and an average of 30 seconds per game was not uncommon. This was due to the more complex network architecture and the more complex action mask, which required a lot of additional calculations.

The final agent was not able to achieve the same level of performance as the Ruehl based agent in more advanced training scenarios. This begs the question of whether this was due to the more complex network and the more restrictive action mask, or whether the final agent simply didn't have enough time to train. The slight increase in score on the reight of 2 suggests that with the same amount of training as the Ruehl based agent, the final agent would be able to at least achieve the same level of performance. However, this is not certain and would require further testing to be sure.

2.3 Conclusion

3 Conclusion

We developed and evaluated a different agents based on Double DQN, a dueling network architecture, engineered spatial and scalar features, action masking, and a staged training curriculum. The curriculum successfully decomposed the task into progressively harder objectives: the agent learned coin collection perfectly in **coin-heaven** and acquired effective crate-destruction behavior in **loot-crate**. In the competitive stages, it also learned to destroy crates, collect coins, and occasionally eliminate opponents while keeping self-destruction rates comparatively low.

In the case of the **ruehl_base_agent**, the agent struggled to achieve consistent success against hard opponents, highlighting the gap between performance in simplified training environments and competitive **classic** games. In particular, the best model against hard opponents achieved a mean score of 3.95 over 20 evaluation games, with a mean of 0.40 opponent eliminations and no completed rounds. Thus, the agent learned meaningful bomb-use and survival-related behavior, but is unable to overcome the enemies counter-strategies consistently.

Possible next steps include evaluation over longer training periods, more systematic hyperparameter tuning, and training against a diverse population of opponents. These extensions would clarify which design choices contribute most strongly to performance and may improve robustness in the full competitive setting.

The other approach we took with the **final_agent** showed that a more complex network and a more restrictive action mask and reward shaping can lead to a faster convergence of the network when only looking at the amount of games played. However, due to the significantly longer training time per game, the final agent was not able to achieve the same level of performance in the same amount of time as the **ruehl_based_agent**. This suggests that the more complex network and the more restrictive action mask and reward shaping are not necessarily better than a simpler approach, but rather a different approach to the same problem, given different constraints.

It can be said that for the task at hand with the given time constraints and the computational resources that we had, the simpler approach of the **ruehl_based_agent** was more effective than the more complex approach of the **final_agent**. However, it is possible that with more time and resources, the **final_agent** could surpass the performance of the **ruehl_based_agent**.

Possible next steps or improvements, to definitely determine the effectiveness of the **final_agent** would be, to train it for a longer period of time, or use a more powerful computer with a gpu to speed up the training time per game.

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